



Midterm Project

Sokoban AI



Course: Artificial Intelligence

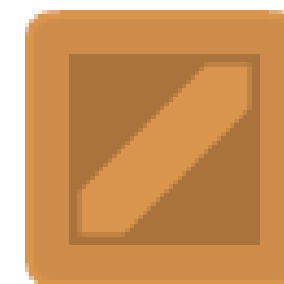
Group: 04



performed by

NGUYỄN ĐỨC HIỆU – 524H0087

NGUYỄN HỮU TÂN – 524H0028



Team Roles & Task Accountability

Student ID	Full Name	Assigned task	Completion
524H0087	Nguyễn Đức Hiệu	Interface design and logic optimization by AgentTan & AgentHieu and testing	100%
524H0028	Nguyễn Hữu Tân	Design A*, UCS, Implement competitive rules and design the analysis.	100%

Diagnostic Checklist: Project Constraints



Core Search Architecture (Task 1) [100%]

- Implement UCS & A* for standard Sokoban
- Propose custom Heuristic (Strictly NO Manhattan/Euclidean)



Competitive AI System (Task 2) [100%]

- Implement Competitive Agents (Agent vs Agent)
- Handle 'C' Box (Neutral on goal) mechanics



Technical Documentation (Task 3) [100%]

- Presentation & Final Report

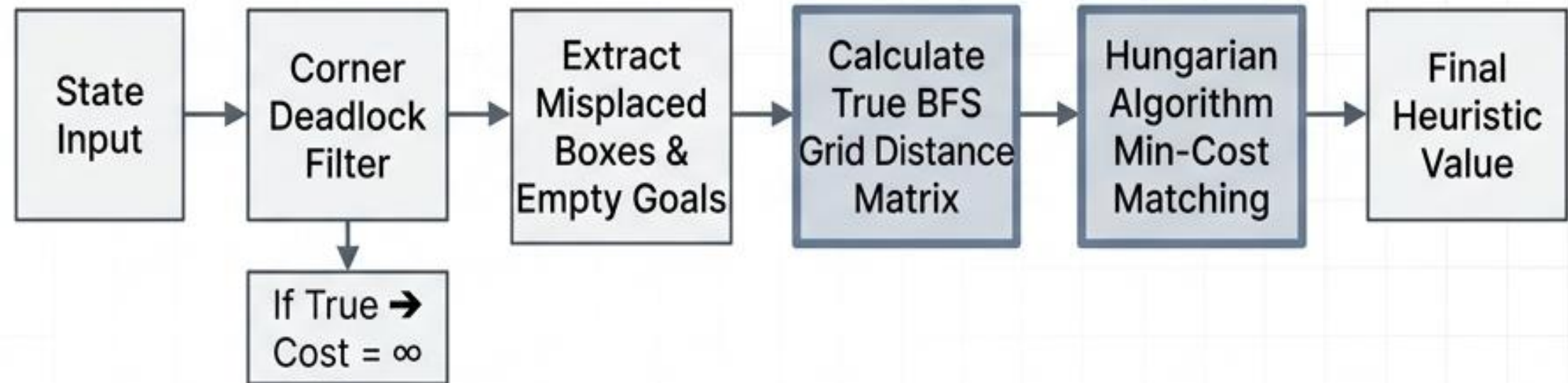
Single-Player Approach: System Pipeline

Uniform Cost Search

Guarantees optimal shortest path via uniform cost, resulting in massive node expansion across the grid.

A* Custom Heuristic Architecture

⚠ Manhattan & Euclidean metrics explicitly banned.

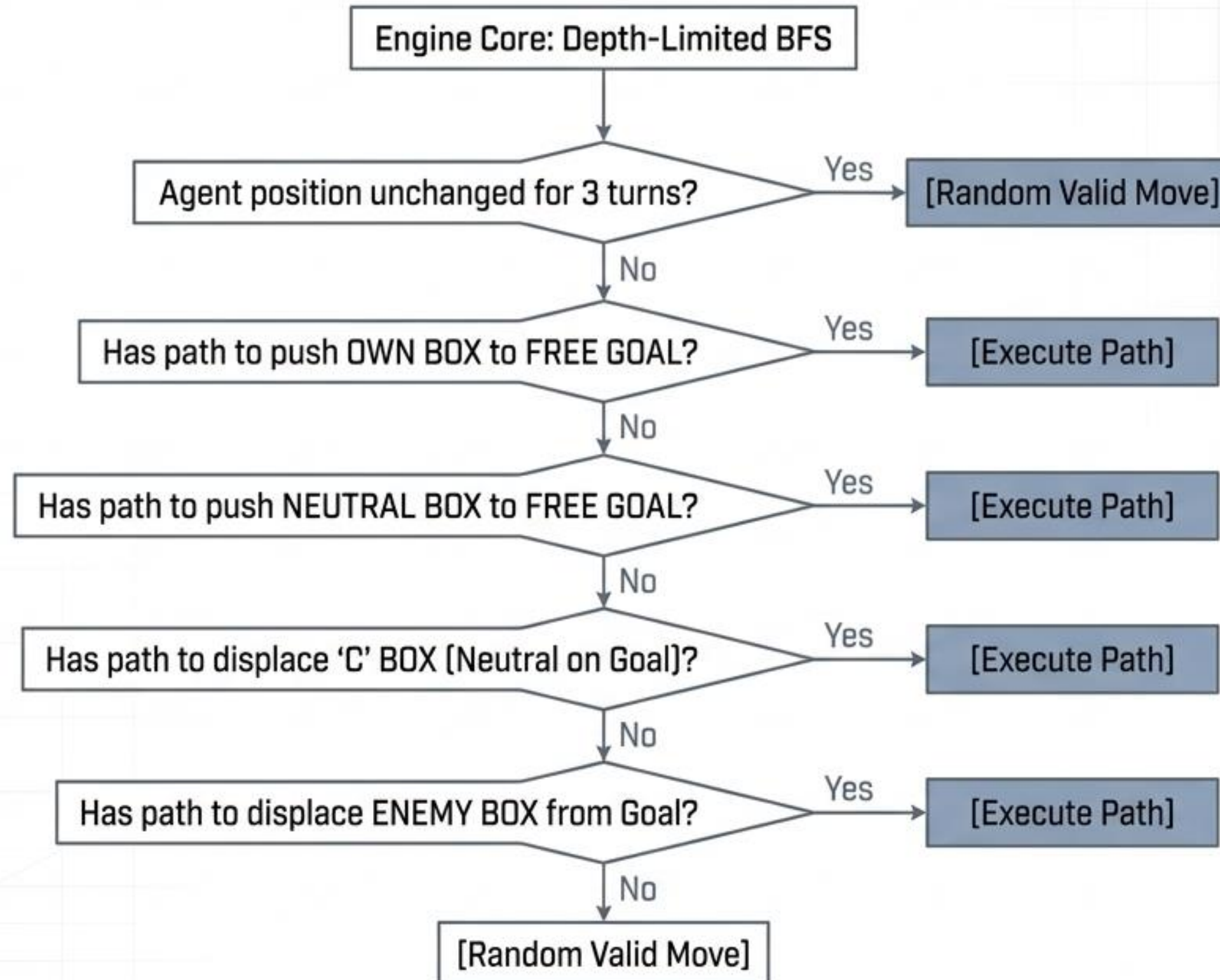


```
Function Heuristic(state):  
    If Corner_Deadlock: Return INFINITY  
    Cost_Matrix = BFS_Distance_Matrix(Boxes, Goals)  
    Return Hungarian_Minimum_Cost(Cost_Matrix)
```

Competitive Multi-Agent Response Logic



< 1000ms
Response Limit



Systemic Advantages & Edge-Case Handling



Strict Rule Compliance

A* heuristic calculates true grid pathfinding, entirely bypassing banned Euclidean/Manhattan metrics while remaining admissible.



Ultra-Fast Execution

Depth-limited local BFS ensures agents react in mere milliseconds, operating safely underneath the strict 1000ms threshold.



Robust Collision Physics

Custom rules engine flawlessly resolves edge-case collisions, including head-to-head blockages and simultaneous pushing attempts.



Strategic 'C' Box Exploitation

Agents dynamically recognize neutral 'C' boxes occupying goals, intelligently ejecting them to claim ownership before scoring.

Calculated Trade-Offs & System Limits

Multi-Agent Myopia

Mechanism: Depth-Limited BFS (capped at 50-60 steps).

Creates a strategic 'fog of war'. On exceptionally large maps, agents may ignore highly optimal boxes located beyond their immediate search horizon in order to maintain sub-second execution speeds.



Heuristic CPU Overhead

Mechanism: Hungarian Algorithm + True BFS distance matrices.

Consumes significantly more CPU cycles per node than simple Manhattan calculations. This overhead guarantees rule compliance but slightly slows node expansion rates on densely complex maps.

Thank You



Thank you teacher for listening!

